11. Constrained least squares

- constrained least squares
- solution of least norm problem
- solution of constrained least squares
- linear quadratic control
- linear quadratic estimation

Constrained least squares

$$\begin{array}{ll} \text{minimize} & \|Ax - b\|^2 \\ \text{subject to} & Cx = d \end{array}$$

- A is an $m \times n$ matrix, b is an m-vector, C is a $p \times n$ matrix, d is a p-vector
- $\|Ax - b\|^2$ is the *objective*, $Cx = d$ are the *constraints*
- we make no assumptions about the shape of A
- in most applications $p < n$ and the equation $Cx = d$ is underdetermined
- goal is to find a solution of $Cx = d$ with smallest objective

Solution

- x is *feasible* if is satisfies the constraints $Cx = d$
- $\hat{x}$ is *optimal* or *solution* or *minimizer* if it is feasible and

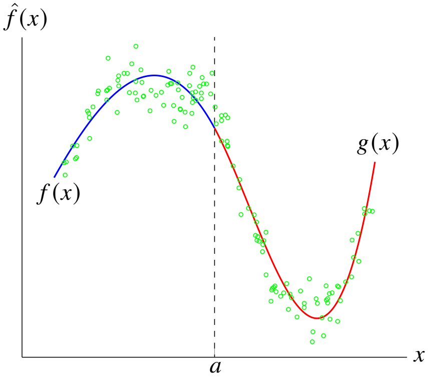
$$\|A\hat{x} - b\|^2 \leq \|Ax - b\|^2 \quad \text{for all feasible } x$$

Example: piecewise-polynomial fitting

- fit two polynomials $f(x)$, $g(x)$ to points $(x_1, y_1), \ldots, (x_N, y_N)$ with

$$f(x_i) \approx y_i \text{ for points } x_i \leq a, \quad g(x_i) \approx y_i \text{ for points } x_i > a$$

- make values and derivatives continuous at point a: $f(a) = g(a)$, $f'(a) = g'(a)$



Example: piecewise-polynomial fitting

Constrained LS formulation

- assume points are numbered so that $x_1, \dots, x_M \le a$ and $x_{M+1}, \dots, x_N > a$:

$$\begin{array}{ll}\text{minimize} & \sum_{i=1}^{M} (f(x_i) - y_i)^2 + \sum_{i=M+1}^{N} (g(x_i) - y_i)^2 \\ \text{subject to} & f(a) = g(a), \quad f'(a) = g'(a)\end{array}$$

- for polynomials $f(x) = \theta_1 + \cdots + \theta_d x^{d-1}$ and $g(x) = \theta_{d+1} + \cdots + \theta_{2d} x^{d-1}$

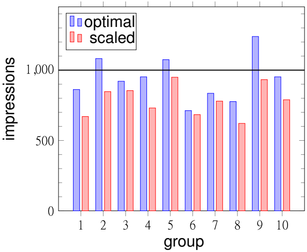
$$A = \left[\begin{array}{cccc|cccc} 1 & x_1 & \cdots & x_1^{d-1} & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 1 & x_M & \cdots & x_M^{d-1} & 0 & 0 & \cdots & 0 \\ \hline 0 & 0 & \cdots & 0 & 1 & x_{M+1} & \cdots & x_{M+1}^{d-1} \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 & 1 & x_N & \cdots & x_N^{d-1} \end{array}\right], \quad b = \left[\begin{array}{c} y_1 \\ \vdots \\ y_M \\ \hline y_{M+1} \\ \vdots \\ y_N \end{array}\right]$$

$$C = \left[\begin{array}{cccc|cccc} 1 & a & \cdots & a^{d-1} & -1 & -a & \cdots & -a^{d-1} \\ 0 & 1 & \cdots & (d-1)a^{d-2} & 0 & -1 & \cdots & -(d-1)a^{d-2} \end{array}\right], \quad d = \left[\begin{array}{c} 0 \\ 0 \end{array}\right]$$

Example: advertising budget allocation

- m demographics groups (audiences), n advertising channels
- b_i is target number of views or impressions for group i
- x_j is amount of advertising purchased in channel j
- A_{ij} is # views in group i per dollar spent on ads in channel j
- $(Ax)_i$ is total number of views in group i
- fixed budget $\mathbf{1}^T x = B$
- constrained LS problem: minimize $\|Ax - b\|^2$ subject to $\mathbf{1}^T x = B$

Example: optimal and scaled (unconstrained) LS solution to satisfy budget



Least norm problem

$$\begin{array}{ll} \text{minimize} & \|x\|^2 \\ \text{subject to} & Cx = d \end{array}$$

- C is a $p \times n$ matrix, d is a p-vector
- the goal is to find the solution of $Cx = d$ with the smallest norm
- a special case of constrained LS with $A = I$ and $b = 0$

Least distance problem: minimizing the distance to a given point $a \neq 0$:

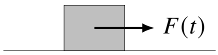
$$\begin{array}{ll} \text{minimize} & \|x - a\|^2 \\ \text{subject to} & Cx = d \end{array}$$

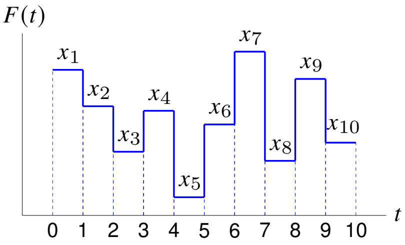
- reduces to least norm problem by a change of variables $y = x - a$

$$\begin{array}{ll} \text{minimize} & \|y\|^2 \\ \text{subject to} & Cy = d - Ca \end{array}$$

- from least norm solution y, we obtain solution $x = y + a$ of first problem

Example: force sequence





- a unit mass with zero initial position and velocity
- we apply piecewise-constant force $F(t)$ during interval $[0, 10)$:

$$F(t) = x_j \quad \text{for } t \in [j-1, j), \quad j = 1, \ldots, 10$$

- final position and velocity at $t = 10$ are given by

$$p^{\text{fin}} = (19/2)x_1 + (17/2)x_2 + (15/2)x_3 + \cdots + (1/2)x_{10}$$
$$v^{\text{fin}} = x_1 + x_2 + \cdots + x_{10}$$

we want to choose a force sequence that results in $p^{\text{fin}} = 1$, $v^{\text{fin}} = 0$

Example: force sequence

there are many solution; we consider two solutions:

1. *bang-bang force:* solutions with only two nonzero elements:

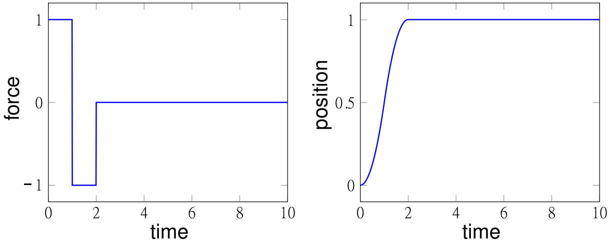
$$x = (1, -1, 0, \ldots, 0), \quad x = (0, 1, -1, \ldots, 0), \ \ldots$$

2. *least norm solution:* smallest force sequence

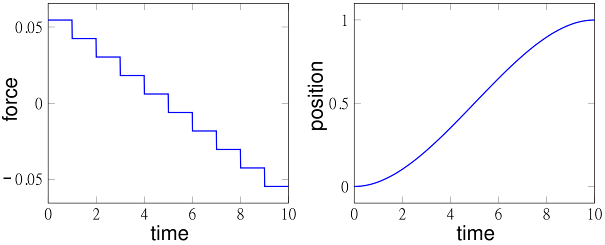
$$\begin{array}{ll} \text{minimize} & \int_0^{10} F(t)^2 dt = \|x\|^2 \\ \text{subject to} & \begin{bmatrix} 19/2 & 17/2 & 15/2 & \cdots & 1/2 \\ 1 & 1 & 1 & \cdots & 1 \end{bmatrix} x = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{array}$$

Example: force sequence

Bang-bang force result



Least norm force result



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Solution of least norm problem

$$\begin{array}{ll} \text{minimize} & \|x\|^2 \\ \text{subject to} & Cx = d \end{array}$$

Assumption: we assume that C has linearly independent rows

- $Cx = d$ has at least one solution for every d
- C is wide or square $(p \leq n)$; if $p < n$ there are infinitely many solutions

Solution of least norm problem

$$\hat{x} = C^T(CC^T)^{-1}d$$

- in other words if $Cx = d$ and $x \neq \hat{x}$, then $\|x\| > \|\hat{x}\|$
- unique solution under the above assumption
- $C^T(CC^T)^{-1} = C^\dagger$ is the pseudo-inverse of C, which is also a right-inverse

Proof

1. we first verify that $\hat{x}$ satisfies the constraints:

$$C\hat{x} = CC^T(CC^T)^{-1}d = d$$

2. next we show that $\|x\| > \|\hat{x}\|$ if $Cx = d$ and $x \neq \hat{x}$

$$\begin{aligned}
\|x\|^2 &= \|\hat{x} + x - \hat{x}\|^2 \\
&= \|\hat{x}\|^2 + 2\hat{x}^T(x - \hat{x}) + \|x - \hat{x}\|^2 \\
&= \|\hat{x}\|^2 + \|x - \hat{x}\|^2 \\
&\geq \|\hat{x}\|^2 \quad \text{with equality only if } x = \hat{x}
\end{aligned}$$

line 3 follows from

$$\hat{x}^T(x - \hat{x}) = d^T(CC^T)^{-1}C(x - \hat{x}) = 0$$

where we used $Cx = C\hat{x} = d$

QR factorization method

using the QR factorization $C^T = QR$ of C^T, we get

$$\begin{aligned} \hat{x} &= C^T(CC^T)^{-1}d \\ &= QR(R^TQ^TQR)^{-1}d \\ &= QR(R^TR)^{-1}d \\ &= QR^{-T}d \end{aligned}$$

Algorithm

1. compute QR factorization $C^T = QR$ ($2p^2n$ flops)
2. solve $R^Tz = d$ by forward substitution (p^2 flops)
3. matrix-vector product $\hat{x} = Qz$ ($2pn$ flops)

Complexity: $2p^2n$ flops

Example

$$C = \begin{bmatrix} 1 & -1 & 1 & 1 \\ 1 & 0 & 1/2 & 1/2 \end{bmatrix}, \quad d = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- QR factorization $C^T = QR$

$$\begin{bmatrix} 1 & 1 \\ -1 & 0 \\ 1 & 1/2 \\ 1 & 1/2 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/\sqrt{2} \\ -1/2 & 1/\sqrt{2} \\ 1/2 & 0 \\ 1/2 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 1/\sqrt{2} \end{bmatrix}$$

- solve $R^T z = d$

$$\begin{bmatrix} 2 & 0 \\ 1 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow z_1 = 0, z_2 = \sqrt{2}$$

- evaluate $\hat{x} = Qz = (1, 1, 0, 0)$

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Assumptions

$$\begin{array}{ll} \text{minimize} & \|Ax - b\|^2 \\ \text{subject to} & Cx = d \end{array}$$

Assumptions

1. the stacked $(m + p) \times n$ matrix
$$\begin{bmatrix} A \\ C \end{bmatrix}$$
has linearly independent columns (left-invertible)

2. $p \times n$ matrix C has linearly independent rows (right-invertible)

assumptions imply that $p \leq n \leq m + p$

Optimality conditions

$$\begin{array}{ll} \text{minimize} & \|Ax - b\|^2 \\ \text{subject to} & Cx = d \end{array}$$

$\hat{x}$ solves the constrained LS problem if and only if there exists a z such that

$$\begin{bmatrix} A^TA & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} \hat{x} \\ z \end{bmatrix} = \begin{bmatrix} A^Tb \\ d \end{bmatrix}$$

- this is a set of $n + p$ linear equations in $n + p$ variables
- equations are also known as *Karush-Kuhn-Tucker* (*KKT*) equations
- matrix on left is called *KKT matrix*

Special cases

- least squares: when $p = 0$, reduces to normal equations $A^TA\hat{x} = A^Tb$
- least norm: when $A = I$, $b = 0$, reduces to $C\hat{x} = d$ and $\hat{x} + C^Tz = 0$

Proof

suppose x satisfies $Cx = d$, and $(\hat{x}, z)$ satisfies optimality conditions, then

$$\begin{aligned}
\|Ax - b\|^2 &= \|A(x - \hat{x}) + A\hat{x} - b\|^2 \\
&= \|A(x - \hat{x})\|^2 + \|A\hat{x} - b\|^2 + 2(x - \hat{x})^T A^T (A\hat{x} - b) \\
&= \|A(x - \hat{x})\|^2 + \|A\hat{x} - b\|^2 - 2(x - \hat{x})^T C^T z \\
&= \|A(x - \hat{x})\|^2 + \|A\hat{x} - b\|^2 \\
&\geq \|A\hat{x} - b\|^2
\end{aligned}$$

- on line 3 we use $A^T A\hat{x} + C^T z = A^T b$; on line 4, $Cx = C\hat{x} = d$
- inequality shows that $\hat{x}$ is optimal
- $\hat{x}$ is the unique optimum because equality holds only if

$$A(x - \hat{x}) = 0, \quad C(x - \hat{x}) = 0 \quad \Longrightarrow \quad x = \hat{x}$$

by the first assumption

Nonsingularity

the KKT matrix

$$\begin{bmatrix} A^TA & C^T \\ C & 0 \end{bmatrix}$$

is nonsingular (invertible) if and only if the two assumptions hold

Proof: if assumptions hold

$$\begin{aligned}
\begin{bmatrix} A^TA & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} &\Longrightarrow x^T(A^TAx + C^Tz) = 0, \quad Cx = 0 \\
&\Longrightarrow \|Ax\|^2 = 0, \quad Cx = 0 \\
&\Longrightarrow Ax = 0, \quad Cx = 0 \\
&\Longrightarrow x = 0 \quad \text{by assumption } 1
\end{aligned}$$

if $x = 0$, we have $C^Tz = -A^TAx = 0$; hence also $z = 0$ by assumption 2

Singularity

if the assumptions do not hold, then the matrix

$$\begin{bmatrix} A^T A & C^T \\ C & 0 \end{bmatrix}$$

is singular

- if assumption 1 does not hold, there exists $x \neq 0$ with $Ax = 0$, $Cx = 0$; then

$$\begin{bmatrix} A^T A & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ 0 \end{bmatrix} = 0$$

- if assumption 2 does not hold there exists a $z \neq 0$ with $C^T z = 0$; then

$$\begin{bmatrix} A^T A & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} 0 \\ z \end{bmatrix} = 0$$

in both cases, this shows that the matrix is singular

Solving KKT equation directly

$$\begin{bmatrix} A^TA & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} A^Tb \\ d \end{bmatrix}$$

Algorithm

1. compute $H = A^TA$ (mn^2 flops)
2. compute $c = A^Tb$ ($2mn$ flops)
3. solve the linear equation
$$\begin{bmatrix} H & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} c \\ d \end{bmatrix}$$
by the LU factorization ($(2/3)(p+n)^3$ flops) or QR factorization ($2(n+p)^3$)

Complexity: $mn^2 + (2/3)(p+n)^3$ flops

Solution by QR factorization

we derive a method that avoid computing gram matrix by using QR factorization

$$\begin{bmatrix} A^TA & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} \hat{x} \\ z \end{bmatrix} = \begin{bmatrix} A^Tb \\ d \end{bmatrix}$$

- multiply 2nd eq. by C^T, add to 1st eq., make change of variables $w = z - d$,

$$\begin{bmatrix} A^TA + C^TC & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} \hat{x} \\ w \end{bmatrix} = \begin{bmatrix} A^Tb \\ d \end{bmatrix}$$

- assumption 1 guarantees $A^TA + C^TC$ is nonsingular with QR factorization:

$$\begin{bmatrix} A \\ C \end{bmatrix} = QR = \begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix} R$$

where Q has orthonormal columns and R nonsingular upper triangular

Solution by QR factorization

substituting $A = Q_1 R$ and $C = Q_2 R$ gives the equation

$$\left[\begin{array}{cc} R^T R & R^T Q_2^T \\ Q_2 R & 0 \end{array}\right] \left[\begin{array}{c} \hat{x} \\ w \end{array}\right] = \left[\begin{array}{c} R^T Q_1^T b \\ d \end{array}\right]$$

- multiply first equation with R^{-T} and make change of variables $y = R\hat{x}$

$$\left[\begin{array}{cc} I & Q_2^T \\ Q_2 & 0 \end{array}\right] \left[\begin{array}{c} y \\ w \end{array}\right] = \left[\begin{array}{c} Q_1^T b \\ d \end{array}\right]$$

- next we note that the matrix $Q_2 = CR^{-1}$ has linearly independent rows:

$$Q_2^T u = R^{-T} C^T u = 0 \quad \Longrightarrow \quad C^T u = 0 \quad \Longrightarrow \quad u = 0$$

  because C has linearly independent rows (assumption 2)

Solution by QR factorization

we use the QR factorization of Q_2^T to solve

$$\begin{bmatrix} I & Q_2^T \\ Q_2 & 0 \end{bmatrix} \begin{bmatrix} y \\ w \end{bmatrix} = \begin{bmatrix} Q_1^T b \\ d \end{bmatrix}$$

- from the 1st block row, $y = Q_1^T b - Q_2^T w$; substitute this in the 2nd row:

$$Q_2 Q_2^T w = Q_2 Q_1^T b - d$$

- we solve this equation for w using the QR factorization $Q_2^T = \tilde{Q}\tilde{R}$:

$$\tilde{R}^T \tilde{R} w = \tilde{R}^T \tilde{Q}^T Q_1^T b - d \quad \Longrightarrow \quad \tilde{R} w = \tilde{Q}^T Q_1^T b - \tilde{R}^{-T} d$$

  we compute $\tilde{R}^{-T} d$ by forward substitution then compute w by back substitution

- from w, we get $y = Q_1^T b - Q_2^T w$ and solve for $\hat{x}$ in $y = R\hat{x}$ via back substitution

Summary of QR factorization method

$$\begin{bmatrix} A^TA + C^TC & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} \hat{x} \\ w \end{bmatrix} = \begin{bmatrix} A^Tb \\ d \end{bmatrix}$$

Algorithm

1. compute the two QR factorizations
$$\begin{bmatrix} A \\ C \end{bmatrix} = \begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix} R \quad \text{and} \quad Q_2^T = \tilde{Q}\tilde{R}$$
2. solve $\tilde{R}^Tu = d$ by forward substitution and compute $c = \tilde{Q}^TQ_1^Tb - u$
3. solve $\tilde{R}w = c$ by back substitution and compute $y = Q_1^Tb - Q_2^Tw$
4. compute $R\hat{x} = y$ by back substitution

Complexity

- $2(m+p)n^2 + 2np^2$ flops (QR factorizations dominates)
- order $(m+p)n^2$ due to assumption $p \le n \le m+p$

Comparison of the two methods

Complexity: LU is slightly more efficient

- LU factorization

$$mn^2 + \frac{2}{3}(p+n)^3 \le mn^2 + \frac{16}{3}n^3 \text{ flops}$$

- QR factorization

$$2(p+m)n^2 + 2np^2 \le 2mn^2 + 4n^3 \text{ flops}$$

upper bounds follow from $p \le n$ (assumption 2)

Stability

- QR factorization method avoids calculation of Gram matrix A^TA
- hence more robust/stable to numerical errors

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Linear quadratic control

Linear dynamical system

$$x_{t+1} = A_t x_t + B_t u_t, \quad y_t = C_t x_t, \quad t = 1, 2, \ldots$$

- n-vector x_t is system *state* (at time t)
- m-vector u_t is system *input* (we control)
- p-vector y_t is system *output*
- x_t, u_t, y_t are typically desired to be small

Objective: choose inputs $u_1, \ldots, u_{T-1}$ that minimizes $J_{\text{output}} + \rho J_{\text{input}}$ with

$$J_{\text{output}} = \|y_1 - y_1^{\text{des}}\|^2 + \cdots + \|y_T - y_T^{\text{des}}\|^2, \quad J_{\text{input}} = \|u_1\|^2 + \cdots + \|u_{T-1}\|^2$$

where y_i^{des} are given desired values (possibly zero)

Constraints

- dynamics constraint
- initial state and (possibly) the final state are specified $x_1 = x^{\text{init}}$, $x_T = x^{\text{des}}$

Linear quadratic control problem

$$\begin{array}{ll} \text{minimize} & \|C_1x_1 - y_1^{\rm des}\|^2 + \cdots + \|C_Tx_T - y_T^{\rm des}\|^2 + \rho\left(\|u_1\|^2 + \cdots + \|u_{T-1}\|^2\right) \\ \text{subject to} & x_{t+1} = A_tx_t + B_tu_t, \quad t = 1, \ldots, T-1 \\ & x_1 = x^{\rm init}, \quad x_T = x^{\rm des} \end{array}$$

variables: $x_1, \ldots, x_T$ and $u_1, \ldots, u_{T-1}$

Constrained least squares formulation

$$\begin{array}{ll} \text{minimize} & \|\tilde{A}z - \tilde{b}\|^2 \\ \text{subject to} & \tilde{C}z = \tilde{d} \end{array}$$

variables: the $(nT + m(T-1))$-vector

$$z = (x_1, \ldots, x_T, u_1, \ldots, u_{T-1})$$

Linear quadratic control problem

Objective function: $\|\tilde{A}z - \tilde{b}\|^2$ with

$$\tilde{A} = \left[\begin{array}{ccc|ccc} C_1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \cdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & C_T & 0 & \cdots & 0 \\ \hline 0 & \cdots & 0 & \sqrt{\rho}I & \cdots & 0 \\ \vdots & \cdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & \sqrt{\rho}I \end{array}\right], \quad \tilde{b} = \left[\begin{array}{c} y_1^{\mathrm{des}} \\ \vdots \\ y_T^{\mathrm{des}} \\ \hline 0 \\ \vdots \\ 0 \end{array}\right]$$

Constraints: $\tilde{C}z = \tilde{d}$ with

$$\tilde{C} = \left[\begin{array}{cccccc|cccc} A_1 & -I & 0 & \cdots & 0 & 0 & B_1 & 0 & \cdots & 0 \\ 0 & A_2 & -I & \cdots & 0 & 0 & 0 & B_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & A_{T-1} & -I & 0 & 0 & \cdots & B_{T-1} \\ \hline I & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 & I & 0 & 0 & \cdots & 0 \end{array}\right], \quad \tilde{d} = \left[\begin{array}{c} 0 \\ 0 \\ \vdots \\ 0 \\ \hline x^{\mathrm{init}} \\ x^{\mathrm{des}} \end{array}\right]$$

Example

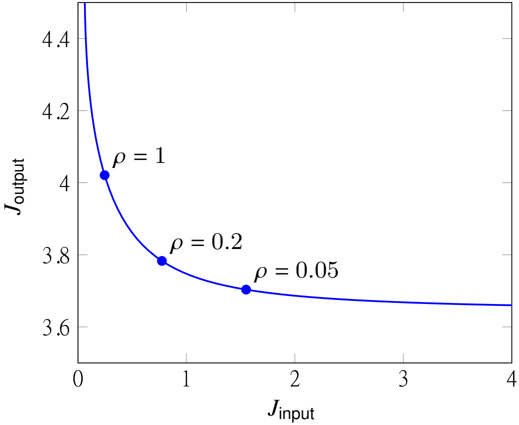
time-invariant system with constant matrices

$$A = \begin{bmatrix} 0.855 & 1.161 & 0.667 \\ 0.015 & 1.073 & 0.053 \\ -0.084 & 0.059 & 1.022 \end{bmatrix}, \quad B = \begin{bmatrix} -0.076 \\ -0.139 \\ 0.342 \end{bmatrix}$$

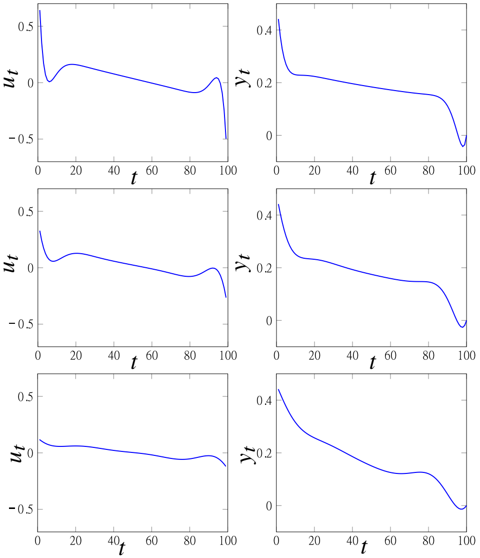
$$C = \begin{bmatrix} 0.218 & -3.597 & -1.683 \end{bmatrix}$$

- $y^{\mathrm{des}} = 0$, $T = 100$
- initial condition $x^{\mathrm{init}} = (0.496, -0.745, 1.394)$
- target or desired final state $x^{\mathrm{des}} = 0$
- input and output have dimension one

Optimal trade-off curve



Three points on the trade-off curve



Linear state feedback control

Linear state feedback

- *linear state feedback control* uses the input

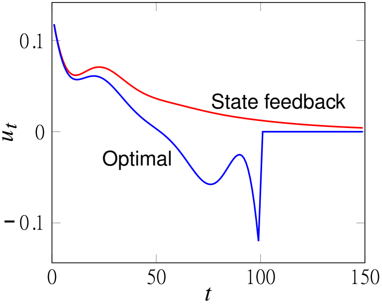
$$u_t = Kx_t, \quad t = 1, 2, \ldots$$

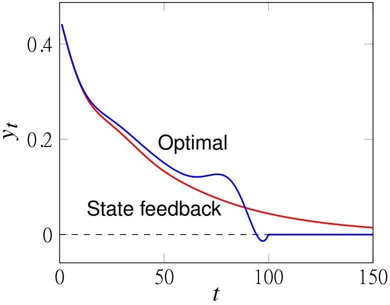
- $K \in \mathbb{R}^{m \times n}$ is the *state feedback gain matrix*
- widely used, especially when x_t should converge to zero and T is not specified

One approach to compute K

- solve the linear quadratic control problem with $x^{\mathrm{des}} = 0$ for (large) T
- solution u_t is a linear function of x^{init}, hence u_1 can be written as $u_1 = Kx^{\mathrm{init}}$
- columns of K can be found by computing u_1 for $x^{\mathrm{init}} = e_1, \ldots, e_n$
- use this K as state feedback gain matrix

Example





- setup of previous example
- blue curve uses optimal linear quadratic control for $T = 100$
- red curve uses simple linear state feedback $u_t = Kx_t$
- optimal choice achieves $y_T = 0$ but linear state feedback makes y_T small only

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State estimation

Linear dynamical system model

$$x_{t+1} = A_t x_t + B_t w_t, \quad y_t = C_t x_t + v_t, \quad t = 1, 2, \ldots$$

- n-vector x_t is *state*
- p-vector y_t is *measurement*
- m-vector w_t is *input* or *process noise*
- p-vector v_t is *measurement noise* or *residual*
- A_t, B_t, C_t are the known dynamics, input, and output matrices

State estimation

- we have measurements $y_1, \ldots, y_T$
- w_t, v_t are unknown, but assumed small
- goal: estimate state sequence $x_1, \ldots, x_T$

Least squares state estimation

$$\begin{array}{ll} \text{minimize} & J_{\text{meas}} + \lambda J_{\text{proc}} \\ \text{subject to} & x_{t+1} = A_t x_t + B_t w_t, \quad t = 1, \ldots, T-1 \end{array}$$

- variables are the states $x_1, \ldots, x_T$ and input noise $w_1, \ldots, w_{T-1}$
- primary objective J_{meas} is sum of squares of measurement residuals:

$$J_{\text{meas}} = \|C_1 x_1 - y_1\|^2 + \cdots + \|C_T x_T - y_T\|^2$$

- secondary objective J_{proc} is sum of squares of process noise

$$J_{\text{proc}} = \|w_1\|^2 + \cdots + \|w_{T-1}\|^2$$

- $\lambda > 0$ is a parameter, trades off measurement and process errors
- similar to control formulation but interpretation is different

Constrained least squares formulation

$$\begin{array}{ll} \text{minimize} & \|C_1x_1 - y_1\|^2 + \cdots + \|C_Tx_T - y_T\|^2 + \lambda\left(\|w_1\|^2 + \cdots + \|w_{T-1}\|^2\right) \\ \text{subject to} & x_{t+1} = A_tx_t + B_tw_t, \quad t = 1, \ldots, T-1 \end{array}$$

- can be written as

$$\begin{array}{ll} \text{minimize} & \|\tilde{A}z - \tilde{b}\|^2 \\ \text{subject to} & \tilde{C}z = \tilde{d} \end{array}$$

- vector z contains the $Tn + (T-1)m$ variables:

$$z = (x_1, \ldots, x_T, w_1, \ldots, w_{T-1})$$

Constrained least squares formulation

$$\tilde{A} = \left[\begin{array}{cccc|ccc} C_1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & C_2 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & C_T & 0 & \cdots & 0 \\ \hline 0 & 0 & \cdots & 0 & \sqrt{\lambda} I & \cdots & 0 \\ \vdots & \vdots & & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & \sqrt{\lambda} I \end{array}\right], \quad \tilde{b} = \left[\begin{array}{c} y_1 \\ y_2 \\ \vdots \\ y_T \\ \hline 0 \\ \vdots \\ 0 \end{array}\right]$$

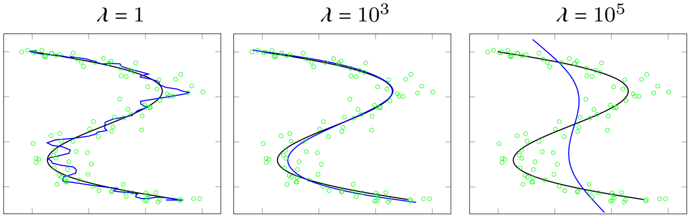
$$\tilde{C} = \left[\begin{array}{cccccc|cccc} A_1 & -I & 0 & \cdots & 0 & 0 & B_1 & 0 & \cdots & 0 \\ 0 & A_2 & -I & \cdots & 0 & 0 & 0 & B_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & A_{T-1} & -I & 0 & 0 & \cdots & B_{T-1} \end{array}\right], \quad \tilde{d} = 0$$

Example

$$A_t = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B_t = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad C_t = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

- simple model of mass moving in a 2-D plane
- $x_t = (p_t, z_t)$: 2-vector p_t is position, 2-vector z_t is the velocity
- $y_t = C_t x_t + w_t$ is noisy measurement of mass position
- $T = 100$

Position estimates



- 100 noisy measurements y_t shown as circles
- solid line is exact position $C_t x_t$
- blue lines show position estimates for three values of λ

References and further readings

- S. Boyd and L. Vandenberghe. *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares.* Cambridge University Press, 2018.

- L. Vandenberghe, *EE133A Lecture Notes,* University of California, Los Angeles.